Oscillators

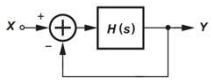
What is an Oscillator?

- An **amplifier** takes an input signal and produces an output signal that is a magnified replica of the input.
- An **oscillator**, on the other hand, is a circuit that *generates* a periodic signal (like a sine wave or a square wave) without requiring a periodic input.
- For example, the clock signal that drives a microprocessor is generated by an on-chip oscillator.

1

The Barkhausen Criteria for Oscillation

- To understand how a circuit can generate a signal on its own, we can think of an oscillator as a feedback amplifier that becomes unstable.
- Consider the standard negative feedback system shown in Figure.



The closed-loop transfer function is given by: $\frac{Y(s)}{X(s)} = \frac{H(s)}{1+H(s)}$

The **gain of this system will approach infinity if the denominator, 1 + H(s), approaches zero**.
This happens at a specific frequency, ω_1, if: **$H(s = j\omega_1) = -1$**

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This single condition can be broken down into two parts, known as the **Barkhausen Criteria**:

1. **Magnitude Condition**: The magnitude of the loop gain must be unity.
   $|H(j\omega_1)| = 1$
2. **Phase Condition**: The total phase shift around the feedback loop must be 360° (or 0°).
   $\angle H(j\omega_1) = 180°$

- The 180° phase shift from the H(s) network, combined with the 180° phase shift from the inverting summing junction (the negative sign in negative feedback), **results in a total phase shift of 360°.**
- This means the signal fed back to the input is perfectly in phase with the original signal, causing positive feedback and sustained oscillation.

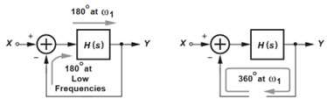


Figure 13.3 (a) Phase shift around an oscillator loop, (b) alternative view.

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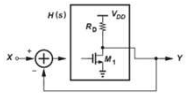
Practical Considerations
Two important questions arise from this theory:

1.Where does the initial signal come from?

- An oscillator has no input. The "seed" for the oscillation is provided by the inherent electronic noise present in any circuit's resistors and transistors.
- This noise contains a wide spectrum of frequencies. The feedback loop selectively amplifies the frequency component that satisfies the Barkhausen criteria, and this component grows until it becomes the steady oscillation.

2.What limits the output amplitude?

- The Barkhausen criteria imply an infinite output.
- In reality, the amplitude is limited by the circuit's physical constraints. As the output signal swing grows, the transistors in the circuit begin to saturate or enter non-linear regions of operation.
- This causes the amplifier's gain to decrease. The amplitude stabilizes when the loop gain $|H(j\omega_1)|$ is reduced to exactly 1.
- For example, in the common-source stage in Figure, as the output swing increases, the transistor M_1 may enter the triode region, reducing its transconductance and thus its gain.



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Startup Condition

To ensure that oscillations begin, the noise signal must be amplified. Therefore, at startup, the loop gain must be **greater than unity**.

$|H(j\omega_1)| > 1$

This is known as the **startup condition**. As the amplitude grows, non-linearities reduce the gain back to 1 for steady oscillation.

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Ring Oscillators

Note: Generally, A ring oscillator is constructed by connecting an odd number of inverting amplifier stages in a ring.

- **A Single-Stage "Oscillator"**:
  - A single common-source (CS) stage (See Figure) provides a 180° phase shift due to its inverting nature.
  - Its single RC pole can provide a maximum of an additional -90° phase shift.
  - The total phase shift is therefore 180° + 90° = 270°, which is insufficient to meet the 360° requirement. Thus, a single-stage loop cannot oscillate.

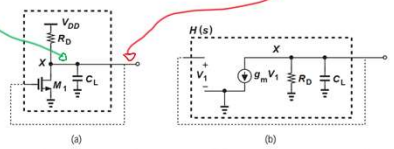


Figure 13.6 (a) Hypothetical oscillator using a single CS stage, (b) equivalent circuit of (a).

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•**A Two-Stage Oscillator: (CS amplifier based)**

- •Cascading two CS stages (See Figure) provides a 360° phase shift at low frequencies (180° from each stage).
- •However, the two poles can provide a maximum of 180° of frequency-dependent phase shift, but this only occurs at an infinite (or High) frequency where the gain is zero.
- •Therefore, we cannot satisfy both Barkhausen criteria simultaneously.

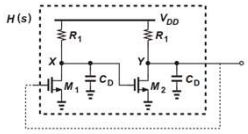


Figure 13.7 Feedback loop using two CS stages.

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A Three-Stage Ring Oscillator:
With three identical CS stages in a loop (See Figure), oscillation is possible.

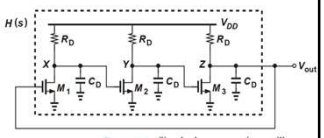


Figure 13.8 Simple three-stage ring oscillator.

•**Phase Condition**:

- ❑ To achieve a total loop phase shift of 360°, the additional phase shift required from the RC networks is 360°.
  - •The there inverting stages
    - •providing 180° x 3 = 540°, or simply -180°,
  - • we need three poles to provide another -180°.
- ❑ For identical stages, each pole must contribute a phase shift of **-60°**. we have

$$-\tan^{-1}(R_D C_D \omega_1) = -60°$$

- **Frequency of Oscillation**: Solving the phase condition gives the frequency of oscillation:

$$\omega_1 = \frac{\sqrt{3}}{R_D C_D}$$

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Startup Condition:

- The transfer function for one stage is

  $H(s) = -gmR_D / (1 + sR_DC_D)$.

- For the loop gain magnitude to be at least 1 at ω_1, the low-frequency gain of *each stage* must be at least **2**.

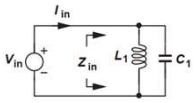
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LC Oscillators

- LC oscillators use inductors (L) and capacitors (C) to form a resonant circuit or "tank" that determines the oscillation frequency.
- They are preferred for high-frequency applications due to their ability to operate faster and with less noise than ring oscillators.

Parallel LC Tanks

An ideal parallel LC tank (See Figure) has an impedance that is infinite at its resonant frequency.



$$\omega_1 = \frac{1}{\sqrt{L_1 C_1}}$$

- At this frequency, the impedance of the inductor (jωL) and the capacitor (1/jωC) are equal and opposite, effectively canceling each other out to create an open circuit.

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- **In practice,** inductors have some series resistance, which represents energy loss.
- This is modeled as a "lossy tank" (See Figure).
- Due to this loss, the impedance at resonance is not infinite but has a finite peak value, Rp.
- An oscillator must include an active component to compensate for this energy loss.

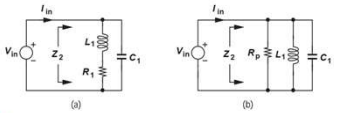


Figure 13.14 (a) Impedance of lossy tank, (b) equivalent circuit of (a).

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Plot the magnitude and phase of $Z_2(s)$ in Fig. as a function of frequency.

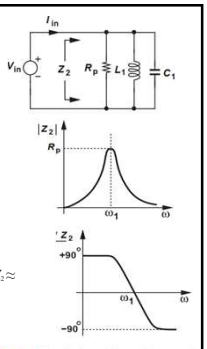
Solution We have

$$Z_2(s) = R_p \| (L_1 s) \| \frac{1}{C_1 s}$$

$$= \frac{R_p L_1 s}{R_p L_1 C_1 s^2 + L_1 s + R_p}$$

At $s = j\omega$,

$$Z_2(j\omega) = \frac{jR_p L_1 \omega}{R_p(1 - L_1 C_1 \omega^2) + jL_1 \omega}$$



- **Case 1:** At $\omega_1 = 1/\sqrt{L_1 C_1}$,
  - we have $Z_2(j\omega_1) = R_p$ and ω_1, $\angle Z_2(j\omega_1) = 0$.
- **Case 2: At very low frequencies**
  - $jL_1\omega$ is very small, dominating the parallel combination, i.e., $Z_2 \approx jL_1\omega$,
- **Case 3: At very high frequencies**
  - $1/(jC_1\omega)$ is very small and hence $Z_2 \approx 1/(jC_1\omega)$.

Figure 13.15 Magnitude and phase of lossy tank.

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Cross-Coupled Oscillator

This is one of the most popular LC oscillator topologies in integrated circuits.

•It can be viewed as two CS amplifiers with LC tank loads, connected in a loop .
•Each CS stage with an LC load provides a 180° phase shift at the resonant frequency ω_1.
•Connecting two such stages in a loop provides the required 360° phase shift for oscillation.

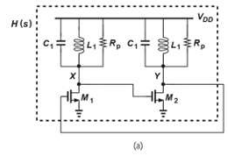


Figure 13.18 (a) Two LC-load CS stages in a loop, (b)

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•The **startup condition** is met if the gain of each stage is at least 1.
•Since the gain at resonance is gm*Rp, the condition for the loop is:

$$(g_m R_p)^2 \geq 1$$

•**A critical issue**
•A critical issue in the above topology is that the bias current of the transistors is poorly defined.
•Since no current mirror or other means of proper biasing are used, the drain currents of M1 and M2 vary with process, supply voltage, and temperature.

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So better circuit is redrawn as follows:
Differential pair with a tail current source, stabilizes the total bias current, making the oscillator's performance robust and predictable, which is essential for integrated circuits

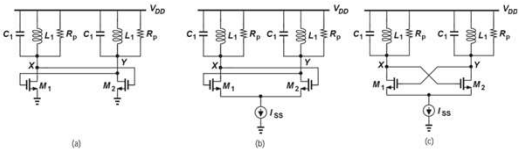


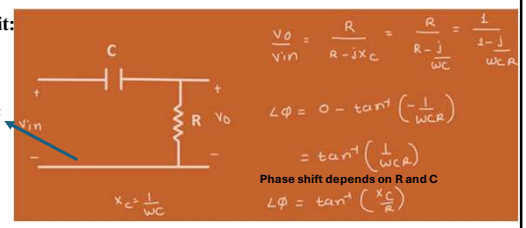
Figure 13.19 Different drawings of the cross-coupled oscillator.

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RC phase-shift oscillator:

Simple RC circuit:

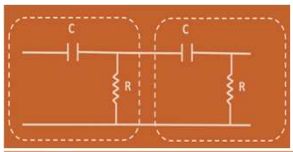
- **High-pass filter**
- **Also known as phase-lead circuit because Vo leads Vin.**



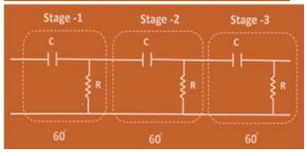
- **For very small values of Xc, the phase shift =0 °**
- **If R=0, then Xc/R=infinite, hence phase shift =90°**

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Cascade connection of RC stages:



- **To get 90° phase shift, the value of R used to be very small or theoretically zero.**
- **But if R=0, the gain =0.**
- **Practically not possible to achieve 180° phase shift by two cascaded RC stages.**
- **So, we require more stages.**



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Phase-Shift Oscillator

- This oscillator uses a single inverting amplifier and a passive RC network to produce the required loop phase shift.

•**Principle**:
  •An inverting amplifier provides 180° of phase shift.
  •The additional 180° needed to satisfy the Barkhausen criteria is provided by a network of resistors and capacitors.
•**Circuit**: A common implementation uses three cascaded RC sections (See Figure).

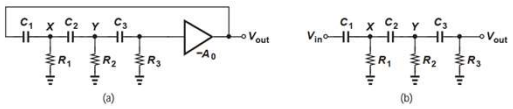


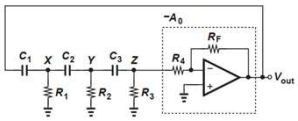
Figure 13.23 (a) Phase shift oscillator, (b) phase shift network.

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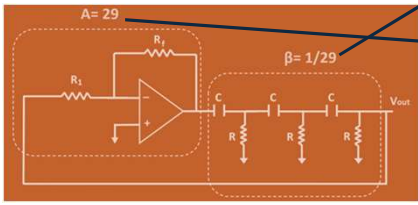
•**Frequency of Oscillation**: The frequency at which the three-stage RC network provides exactly 180° of phase shift is:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

•**Startup Condition**:
•At this frequency, the RC network also attenuates the signal by a factor of 29.
•Therefore, to achieve a loop gain of at least 1, the inverting amplifier must have a gain of at least **29**.



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Attenuation provided by the feedback circuit is 1/29.
So, the Gain provided by the opamp should be equal to 29 to get a unity loop gain.

$$f = \frac{1}{2\pi RC\sqrt{2N}}$$

For 3 stage RC network:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

At this frequency only unity loop gain and zero phase shift is achieved for this oscillator.

- **Only at one particular frequency, the op amp circuit will provide unity loop gain.**
- **The overall phase shift of the circuit will be equal to zero at that one particular frequency only.**

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Wien-Bridge Oscillator

The Wien-bridge oscillator is known for generating high-quality sine waves and also uses a single amplifier stage.

•**Principle**:
•Unlike the phase-shift oscillator, the Wien-bridge uses a **non-inverting** amplifier (0° phase shift).
•Therefore, the passive feedback network must also have a **0° phase shift** at the desired frequency.
•**Circuit**: The passive network is a lead-lag network (See Figure).

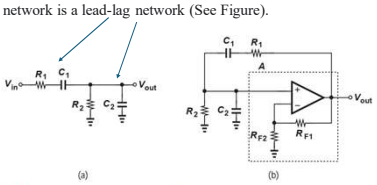
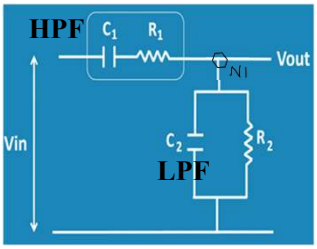


Figure 13.26 (a) Phase shift network, (b) Wien-bridge oscillator.

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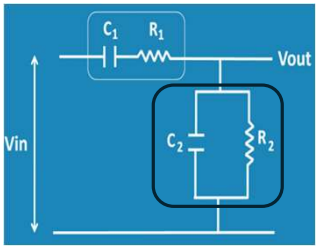
Wein Bridge Oscillator:



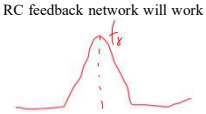
- At low frequencies, C1 acts as an open circuit and does not pass low-frequency signals.
- At high frequencies, C1 provides very low impedance and allows high frequency components.
- At low frequencies, C2 acts as an open circuit so that output appears across node N1.
- At high frequencies, C2 provides very low impedance and o/p gets short circuited to ground terminal and acts as LPF.

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Wein Bridge Oscillator:



- This RC network does not allow low as well as high frequencies.
- But at one particular frequency, the o/p of the circuit will be maximum, known as the resonant frequency (f_R).
- So, in a way, this RC feedback network will work as a notch filter.

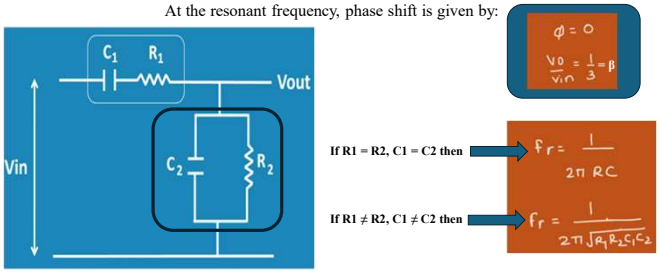


Only at one particular frequency, output will be maximum, and the rest of the frequencies, output will be minimum.

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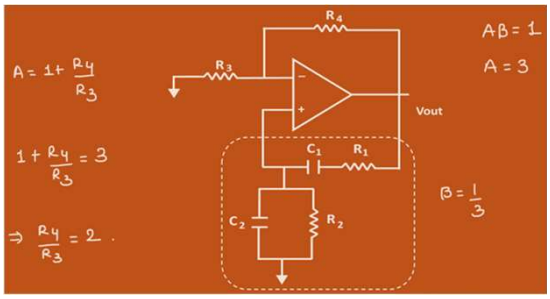
Wein Bridge Oscillator:

At the resonant frequency, phase shift is given by:



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Wein Bridge Oscillator:



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Wein Bridge Oscillator:

Lets design a Wein bridge oscillator of frequency = 10KHz

$R_1 = R_2$

$C_1 = C_2$

$f = 10kHz$

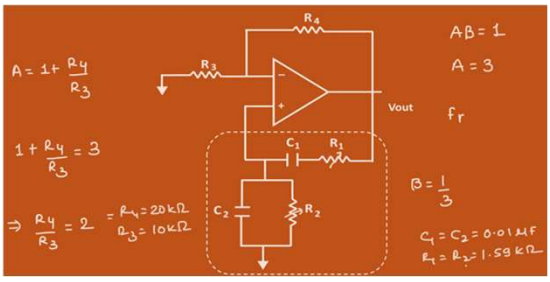
$C = 0.01\mu F$

$f_r = \frac{1}{2\pi RC}$

$R = \frac{1}{2\pi \times f \times C} = \frac{1}{2\pi \times 10^4 \times 10^{-8}} \simeq 1.59k\Omega$

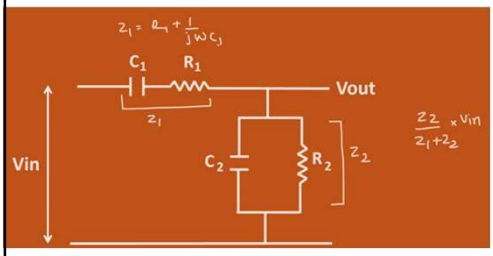
26

Wein Bridge Oscillator:



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Wein Bridge Oscillator:



$\frac{V_o}{V_{in}} = \frac{Z_2}{Z_1+Z_2}$

$Z_2 = R_2 \| \left(\frac{1}{j\omega C_2}\right) = \frac{R_2}{1+j\omega R_2 C_2}$

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$\frac{V_o}{V_{in}} = \frac{Z_2}{Z_1+Z_2}$

$Z_2 = R_2 \| \left(\frac{1}{j\omega C_2}\right) = \frac{R_2}{1+j\omega R_2 C_2}$

$\frac{V_o}{V_{in}} = \frac{\frac{R_2}{1+j\omega R_2 C_2}}{R_1 + \frac{1}{j\omega C_1} + \frac{R_2}{1+j\omega C_2 R_2}}$

$\frac{V_o}{V_{in}} = \frac{R_2(j\omega C_1)}{R_1(j\omega C_1)(1+j\omega C_2 R_2) + 1 + j\omega C_2 R_2 + j\omega R_2 C_1}$

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Wein Bridge Oscillator:

$\frac{V_o}{V_{in}} = \frac{j\omega R_2 C_1}{\underbrace{1-\omega^2 R_1 R_2 C_1 C_2}_{0} + j\omega[R_1 C_1 + R_2 C_2 + R_2 C_1]}$

$\omega^2 R_1 R_2 C_1 C_2 = 1$

$\Rightarrow \omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}} \Rightarrow f = \frac{1}{2\pi \times \sqrt{R_1 R_2 C_1 C_2}} \Rightarrow f = \frac{1}{2\pi RC}$

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Wein Bridge Oscillator

$$\frac{V_o}{V_{in}} = \frac{R_2C_1}{R_1C_1 + R_2C_2 + R_2C_1} = \beta \qquad \beta = \frac{1}{3}$$

$$A\beta = 1$$

$$A = \frac{1}{\beta} = \frac{R_1C_1 + R_2C_2 + R_2C_1}{R_2C_1} = \frac{R_1}{R_2} + \frac{C_2}{C_1} + 1$$

$$A = 1 + \frac{R_4}{R_3} = \frac{R_1}{R_2} + \frac{C_2}{C_1} + 1$$

$$\Rightarrow \frac{R_4}{R_3} = \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

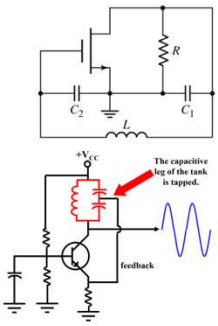
$R_1 = R_2$, $C_1 = C_2$

$$\Rightarrow \frac{R_4}{R_3} = 2$$

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Colpitts oscillator

- The Colpitts oscillator is a type of oscillator that uses an LC circuit in the feed-back loop.
- The feedback network is made up of a pair of tapped capacitors (C1and C2) and an inductor L to produce a feedback necessary for oscillations.
- The output voltage is developed across C_1.
- The feedback voltage is developed across C2.



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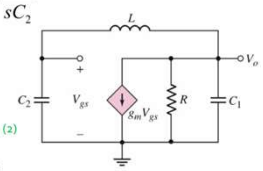
Colpitts oscillator

KCL at the output node:

$$\frac{V_o}{\frac{1}{sC_1}} + \frac{V_o}{R} + g_m V_{gs} + \frac{V_o}{sL + \frac{1}{sC_2}} = 0 \quad \text{- Eq (1)}$$

voltage divider produces:

$$V_{gs} = \left(\frac{\frac{1}{sC_2}}{\frac{1}{sC_2} + sL}\right) \bullet V_o \quad \text{- Eq (2)}$$



substitute eq(2) into eq(1):

$$V_o\left[g_m + sC_2 + \left(1 + s^2LC_2\right)\left(\frac{1}{R} + sC_1\right)\right] = 0$$

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Colpitts oscillator

Assume that oscillation has started, then Vo≠o

$$s^3LC_1C_2 + \frac{s^2LC_2}{R} + s(C_1 + C_2) + \left(g_m + \frac{1}{R}\right) = 0$$

Let s=jω

$$\left(g_m + \frac{1}{R} + \frac{\omega^2LC_2}{R}\right) + j\omega\left[(C_1 + C_2) - \omega^2LC_1C_2\right] = 0$$

both real & imaginary component must be zero

- Imaginary component:

$$\omega_o = \frac{1}{\sqrt{L\left(\frac{C_1C_2}{C_1 + C_2}\right)}} \quad \text{- Eq (3)}$$

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Colpitts oscillator

both real & imaginary component must be zero

- Imaginary component:

$$\frac{\omega^2LC_2}{R} = g_m + \frac{1}{R} \quad \text{- Eq (4)}$$

Combining Eq(3) and Eq(4):

$$\frac{C_2}{C_1} = g_mR$$

to initiate oscillations spontaneously:

$$g_mR > \left(\frac{C_2}{C_1}\right)$$

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